Data-Backed Journal-Style Replication of *Position and Differentiation of Firms in Technology Space*

Codex journal-replication workflow

2026-04-04

Abstract

This manuscript expands the local replication folder for Arts, Cassiman, and Hou (2023) into a long journal-style paper using only processed datasets already present in the workspace. The current local batch contains 5,192 observations across 443 entities, and the batch headline estimate is -0.1935 for tech_diff with p = 0.7841. The local headline coefficient on technology differentiation is negative and statistically weak after firm and year effects are imposed. The draft imitates the structure of the source article while documenting the exact places where local replication decisions depart from direct source facts.

Keywords: technology differentiation; firm performance; patent portfolios; local calculated replication

1 Paper-First Reading Protocol

The workflow for this manuscript is intentionally ordered. The source paper is read first, the required empirical objects are identified second, the local files are verified third, and only then is the replication paper written from the checked evidence chain.

Step	Action	Evidence
1	Read the original paper before touching the local estimates.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/arts-et-al-2023-position-and-differentiation-of-firms-in-technology-space.txt
2	Extract the source paper's theoretical logic, sample definition, variable definitions, and estimand.	Position and Differentiation of Firms in Technology Space

Step	Action	Evidence
3	Match each required empirical object to a local executable file and verify that the file exists.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_analysis_dataset.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_model_coefficients.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_sample_summary.csv
4	Write the replication paper only from the checked local files and calculated outputs.	The manuscript, markdown notes, and PDF are generated from the verified file set.

Ordered protocol used for this replication paper.

2 Required Data and Local File Manifest

This section answers the basic replication question of where the data are. Each empirical object is mapped to a concrete local file before the estimates are interpreted. Workspace root: C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions

Stage	NeededFor	FilePath	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	patent literature/arts-et-al-2023-position-and-differentiation-of-firms-in-technology-space.txt	Yes	Direct source
Analysis panel	Estimate the local model and regenerate descriptive statistics	patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_analysis_dataset.csv	Yes	Executable local dataset
Coefficient export	Recover the reported local coefficients and p-values	patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_model_coefficients.csv	Yes	Executable local result file

Stage	NeededFor	FilePath	Exists	Status
Sample summary	Audit sample construction and raw grouped moments	patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_sample_summary.csv	Yes	Executable local result file
Fit summary	Check model diagnostics when a fit file is available	patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_model_fit.csv	Yes	Executable local result file

Exact local files used by the manuscript.

3 Introduction

The source paper asks whether technologically differentiated firms perform better than their peers. The point of the present draft is not to pretend that every original source file has been recovered. It is to turn the executable local evidence into a paper that preserves theory, data architecture, variable mapping, and empirical claims in one place. The current local batch contains 5,192 observations across 443 entities, and the batch headline estimate is -0.1935 for tech_diff with p = 0.7841. The local headline coefficient on technology differentiation is negative and statistically weak after firm and year effects are imposed.

4 Research Positioning and Contribution

The replication matters because it tests whether a central strategy result about technological differentiation survives when the semantic layer is stripped back to a fully executable local panel. The paper's contribution is a transparent calculated analogue: it preserves the peer-comparison logic and performance outcome while making the measurement substitution explicit. The writing stance follows a conservative paper-first workflow: the source paper defines the theoretical target, the local files define the executable boundary, and the replication paper names the distance between the two rather than smoothing it over.

5 Original Paper, Theory, and Published Estimand

The focal source article is [1]. The source paper is Position and Differentiation of Firms in Technology Space by Sam Arts, Bruno Cassiman, and Jianan Hou in Management Science . The published paper builds technology-space measures from patent text and studies differentiation among U.S. public firms. The main outcome is firm performance, especially Tobin's Q. Technology differentiation is defined relative to technologically similar peers in the same year. A useful replication discipline is to keep source facts and replication decisions separate. Source facts describe what the paper says. Replication decisions describe what the local executable workflow must do because it is limited to processed datasets already present in the workspace.

- The published paper builds technology-space measures from patent text and studies differentiation among U.S. public firms.
- The main outcome is firm performance, especially Tobin's Q.
- Technology differentiation is defined relative to technologically similar peers in the same year.

DirectSourceFact	LocalReplicationDecision
The source paper uses semantic patent content.	The local rerun uses processed primary-USPC class portfolios.
Tobin's Q is a central outcome.	The local panel keeps Tobin's Q because it is fully reconstructible.
Differentiation is relative to peers.	Peer comparison is preserved with a class-based similarity measure.

Direct source facts are kept separate from local replication decisions.

6 Decision Register

The table below makes the major design choices explicit before any coefficient is interpreted.

DecisionPoint	LocalSpecification
Research question	The source paper asks whether technologically differentiated firms perform better than their peers.
Unit of analysis	Firm-year
Sample frame	Processed public-firm GVKEY-year panel with patent portfolios and financial controls.
Time window	1991 to 2015 in the local executable panel.

DecisionPoint	LocalSpecification
Outcome definition	Winsorized Tobin's Q at the firm-year level.
Main explanatory design	Firm and year fixed-effects regression of Tobin's Q on local technology differentiation and controls.
Controls and fixed effects	Financial controls plus firm and year fixed effects.
Exclusion or substitution logic	Original semantic embedding inputs are unavailable locally, so the executable rerun substitutes class-based portfolio similarity and keeps only observations supported by the processed panel.

Decision register for the local replication design.

7 Local Data Inventory and Sample Reconstruction

The local rerun replaces the paper's text-semantic portfolio construction with a class-based five-year rolling USPC portfolio built from processed patent-firm assignments. Every number in this manuscript is backed by local files in patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space or by the explicitly referenced processed regression panel.

- DISCERN-linked firm-patent assignments with unambiguous GVKEYs.
- PatentView-derived utility-patent and primary-USPC class information.

- Compustat annual fundamentals used to reconstruct Tobin's Q and controls.

File	Role
arts_2023_analysis_dataset.csv	Firm-year analysis panel
arts_2023_model_coefficients.csv	Estimated coefficients
arts_2023_model_fit.csv	Model fit statistics
arts_2023_sample_summary.csv	Sample reconstruction summary

Local executable files used by this paper.

Statistic	Value
rows	5192.0000
unique_firms	443.0000
year_min	1991.0000
year_max	2015.0000
mean_tobin_q_w	2.1771
mean_tech_diff	0.6682
sd_tech_diff	0.1733

Sample reconstruction table from the local results folder.

8 Variable Mapping, Derivations, and Replication Decisions

The variable map is the bridge between the source article and the executable local workflow.

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Technology portfolio	Five-year rolling primary-USPC portfolio	portfolio_patents; class_count; class_hhi	Aggregate utility patents by GVKEY and primary USPC class over t-5 to t-1.
Technology differentiation	One minus peer similarity	tech_diff	Compute one minus average similarity to close peers.
Peer benchmark	Nearest peers in year t	neighbor_similarity	Use local class-based portfolios to define the similarity set.
Firm performance	Winsorized Tobin's Q	tobin_q_w	Reconstruct Tobin's Q from processed Compustat variables and winsorize.
Financial controls	Assets, leverage, cash, R&D, portfolio size	log_assets; leverage; cash_ratio; rd_intensity; log1p(portfolio_patents)	Carry forward the financial controls already built in the local script.

Variable mapping from source constructs to local measures. A firm and year fixed-effects regression of winsorized Tobin's Q on local technology differentiation and financial controls.

$$Q_{it}^{(w)} = \beta_1 \text{TechDiff}_{it} + \beta_2 X_{it} + \alpha_i + \gamma_t + \varepsilon_{it}. \tag{1}$$

9 Empirical Design and Identification Limits

The local design remains strongest when it preserves the original unit of analysis and interaction logic, even if some measurement layers are reconstructed with defensible analogues rather than identical source files.

- The exact text-embedding pipeline from the source paper is not stored locally.
- The local differentiation measure is therefore class-based rather than text-semantic.
- Comparisons to the published estimates should be read as inferential rather than byte-identical.

10 Main Results

Technology differentiation is estimated at -0.1935 with p = 0.7841 in model arts_fe. Log assets is estimated at -0.5161 with p = 0.0017 in model arts_fe. Leverage is estimated at -0.6522 with p = 0.0315 in model arts_fe. Cash ratio is estimated at 1.5652 with p = 0.0226 in model arts_fe. R&D intensity is estimated at 4.2903 with p = 0.0093 in model arts_fe. Log(1 + portfolio patents) is estimated at -0.2660 with p = 0.0021 in model arts_fe. Class count is estimated at 0.0043 with p = 0.0851 in model arts_fe. Class concentration is estimated at 0.4918 with p = 0.1094 in model arts_fe.

n	ll	aic	bic	wr2	model
4669.0000	-6268.8113	13389.6226	16136.7689	0.1440	arts_fe

Model fit statistics from the local results folder.

model	Term	estimate	std_error	p_value
arts_fe	Technology differentiation	-0.1935	0.6974	0.7841
arts_fe	Log assets	-0.5161	0.1442	0.0017
arts_fe	Leverage	-0.6522	0.2840	0.0315
arts_fe	Cash ratio	1.5652	0.6384	0.0226
arts_fe	R&D intensity	4.2903	1.5054	0.0093
arts_fe	Log(1 + portfolio patents)	-0.2660	0.0764	0.0021
arts_fe	Class count	0.0043	0.0024	0.0851
arts_fe	Class concentration	0.4918	0.2947	0.1094

Key coefficient table used in the main text.

Term	Model	Estimate	PValue	Signal	Reading
Technology differentiation	arts_fe	-0.1935	0.7841	Weak	Negative estimate with p = 0.7841.
Log assets	arts_fe	-0.5161	0.0017	Strong	Negative estimate with p = 0.0017.
Leverage	arts_fe	-0.6522	0.0315	Moderate	Negative estimate with p = 0.0315.

Term	Model	Estimate	PValue	Signal	Reading
Cash ratio	arts_fe	1.5652	0.0226	Moderate	Positive estimate with p = 0.0226.
R&D intensity	arts_fe	4.2903	0.0093	Strong	Positive estimate with p = 0.0093.
Log(1 + portfolio patents)	arts_fe	-0.2660	0.0021	Strong	Negative estimate with p = 0.0021.
Class count	arts_fe	0.0043	0.0851	Suggestive	Positive estimate with p = 0.0851.
Class concentration	arts_fe	0.4918	0.1094	Weak	Positive estimate with p = 0.1094.

Interpretive summary of the main coefficients.

Variable	N	Mean	SD	P25	Median	P75	Min	M
Winsorized Tobin's Q	5192.0000	2.1771	1.5081	1.3189	1.7253	2.4637	0.4751	23.07
Technology differentiation	5192.0000	0.6682	0.1733	0.5576	0.6935	0.8000	0.0866	0.99

Variable	N	Mean	SD	P25	Median	P75	Min	M
Neighbor similarity	5192.0000	0.3318	0.1733	0.2000	0.3065	0.4424	0.0050	0.91
Portfolio patents	5192.0000	552.7049	1483.5605	29.0000	129.0000	450.0000	1.0000	33760.0
Class count	5192.0000	41.7427	42.9175	10.0000	28.0000	57.0000	1.0000	250.00
Class concentration	5192.0000	0.2126	0.2196	0.0714	0.1347	0.2653	0.0163	1.00
Log assets	5192.0000	8.8404	1.3569	7.9058	8.7190	9.6639	4.3114	13.58
Leverage	5192.0000	0.5877	0.2216	0.4541	0.5836	0.7112	0.0317	2.93
Cash ratio	5192.0000	0.1348	0.1468	0.0280	0.0802	0.1909	0.0000	0.85
R&D intensity	4685.0000	0.0519	0.0540	0.0146	0.0327	0.0741	0.0000	0.67

Descriptive statistics for the main variables.

Quantile	N	MeanRegressor	MeanOutcome
5.0000	1039.0000	0.8832	1.9569
4.0000	1038.0000	0.7780	1.9779
3.0000	1038.0000	0.6925	1.9966
2.0000	1038.0000	0.5863	2.0965
1.0000	1039.0000	0.4011	2.8572

Quantile summary linking the main regressor to the local outcome.

11 Claim-Evidence Audit

This audit prevents the paper from blurring together original-paper statements, local estimates, and replication judgments.

Claim	EvidenceType	EvidenceFile	AuditNote
The source paper studies: The source paper asks whether technologically differentiated firms perform better than their peers.	Direct source fact	patent literature/arts-et-al-2023-position-and-differentiation-of-firms-in-technology-space.txt	Recovered from the source article before local writing begins.
The local executable panel contains 5,192 observations across 443 entities.	Local calculated fact	patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_analysis_dataset.csv — patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_sample_summary.csv	Backed by the verified analysis panel and sample audit.
The local headline estimate is -0.1935 for tech_diff with p = 0.7841.	Local calculated fact	patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_model_coefficients.csv	Backed by the exported coefficient file used in the paper.

Claim	EvidenceType	EvidenceFile	AuditNote
The manuscript uses local substitutions or analogues wherever the original data architecture is not fully recoverable.	Replication decision	Source-vs-decision table and limitations section	Used only when the original pipeline cannot be matched byte-for-byte from current local files.

Claim-evidence audit for the current manuscript.

12 Comparison to the Published Paper and Remaining Gaps

The source paper reports a positive performance relationship, whereas the local executable analogue produces a negative and imprecise coefficient. That gap is useful evidence about how much the published result may depend on the original semantic measurement layer and sample construction. The honest replication standard is to describe the gap without smoothing it over. A useful local paper tells the next researcher exactly which parts of the original design are recoverable now and which parts remain approximated.

13 Discussion and Conclusion

The manuscript turns a replication folder into a paper that can be uploaded, cited internally, and rerun later without guessing how the evidence was assembled. For Arts, Cassiman, and Hou (2023) the bottom line is: The local headline coefficient on technology differentiation is negative and statistically weak after firm and year effects are imposed.

A Appendix A: Source Inventory

Stage	NeededFor	FilePath	Exists	Status
Source article	Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices	patent literature/arts-et-al-2023-position-and-differentiation-of-firms-in-technology-space.txt	Yes	Direct source
Analysis panel	Estimate the local model and regenerate descriptive statistics	patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_analysis_dataset.csv	Yes	Executable local dataset
Coefficient export	Recover the reported local coefficients and p-values	patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_model_coefficients.csv	Yes	Executable local result file

Stage	NeededFor	FilePath	Exists	Status
Sample summary	Audit sample construction and raw grouped moments	patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_sample_summary.csv	Yes	Executable local result file
Fit summary	Check model diagnostics when a fit file is available	patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_model_fit.csv	Yes	Executable local result file

Source and local file manifest for the manuscript.

B Appendix B: Reading Protocol

Step	Action	Evidence
1	Read the original paper before touching the local estimates.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/arts-et-al-2023-position-and-differentiation-of-firms-in-technology-space.txt
2	Extract the source paper's theoretical logic, sample definition, variable definitions, and estimand.	Position and Differentiation of Firms in Technology Space

Step	Action	Evidence
3	Match each required empirical object to a local executable file and verify that the file exists.	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_analysis_dataset.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_model_coefficients.csv — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_sample_summary.csv
4	Write the replication paper only from the checked local files and calculated outputs.	The manuscript, markdown notes, and PDF are generated from the verified file set.

Paper-first protocol followed before writing the manuscript.

C Appendix C: Decision Register

DecisionPoint	LocalSpecification
Research question	The source paper asks whether technologically differentiated firms perform better than their peers.
Unit of analysis	Firm-year
Sample frame	Processed public-firm GVKEY-year panel with patent portfolios and financial controls.
Time window	1991 to 2015 in the local executable panel.
Outcome definition	Winsorized Tobin's Q at the firm-year level.
Main explanatory design	Firm and year fixed-effects regression of Tobin's Q on local technology differentiation and controls.
Controls and fixed effects	Financial controls plus firm and year fixed effects.

DecisionPoint	LocalSpecification
Exclusion or substitution logic	Original semantic embedding inputs are unavailable locally, so the executable rerun substitutes class-based portfolio similarity and keeps only observations supported by the processed panel.

Expanded decision register for the manuscript.

D Appendix D: Sample Reconstruction

Statistic	Value
rows	5192.0000
unique_firms	443.0000
year_min	1991.0000
year_max	2015.0000
mean_tobin_q_w	2.1771
mean_tech_diff	0.6682
sd_tech_diff	0.1733

Detailed sample reconstruction table.

E Appendix E: Variable Mapping

OriginalConstruct	LocalMeasure	ExactVariable	Derivation
Technology portfolio	Five-year rolling primary-USPC portfolio	portfolio_patents; class_count; class_hhi	Aggregate utility patents by GVKEY and primary USPC class over t-5 to t-1.
Technology differentiation	One minus peer similarity	tech_diff	Compute one minus average similarity to close peers.
Peer benchmark	Nearest peers in year t	neighbor_similarity	Use local class-based portfolios to define the similarity set.
Firm performance	Winsorized Tobin's Q	tobin_q_w	Reconstruct Tobin's Q from processed Compustat variables and winsorize.
Financial controls	Assets, leverage, cash, R&D, portfolio size	log_assets; leverage; cash_ratio; rd_intensity; log1p(portfolio_patents)	Carry forward the financial controls already built in the local script.

Detailed construct mapping.

F Appendix F: Descriptive Statistics

Variable	N	Mean	SD	P25	Median	P75	Min	M
Winsorized Tobin's Q	5192.0000	2.1771	1.5081	1.3189	1.7253	2.4637	0.4751	23.07
Technology differentiation	5192.0000	0.6682	0.1733	0.5576	0.6935	0.8000	0.0866	0.99
Neighbor similarity	5192.0000	0.3318	0.1733	0.2000	0.3065	0.4424	0.0050	0.91
Portfolio patents	5192.0000	552.7049	1483.5605	29.0000	129.0000	450.0000	1.0000	33760.0
Class count	5192.0000	41.7427	42.9175	10.0000	28.0000	57.0000	1.0000	250.00
Class concentration	5192.0000	0.2126	0.2196	0.0714	0.1347	0.2653	0.0163	1.00
Log assets	5192.0000	8.8404	1.3569	7.9058	8.7190	9.6639	4.3114	13.58
Leverage	5192.0000	0.5877	0.2216	0.4541	0.5836	0.7112	0.0317	2.93
Cash ratio	5192.0000	0.1348	0.1468	0.0280	0.0802	0.1909	0.0000	0.85
R&D intensity	4685.0000	0.0519	0.0540	0.0146	0.0327	0.0741	0.0000	0.67

Descriptive statistics appendix.

G Appendix G: Correlation Matrix

Variable	Winsorized Tobin's Q	Technology differentiation	Neighbor similarity	Portfolio patents	Class count	Class concentration	Log assets	Leverage	Cash ratio
Winsorized Tobin's Q	1.000	-0.214	0.214	-0.029	-0.144	0.213	-0.097	-0.245	0.357
Technology differentiation	-0.214	1.000	-1.000	-0.221	-0.153	0.106	-0.224	0.125	-0.298
Neighbor similarity	0.214	-1.000	1.000	0.221	0.153	-0.106	0.224	-0.125	0.298
Portfolio patents	-0.029	-0.221	0.221	1.000	0.627	-0.200	0.414	0.037	0.044
Class count	-0.144	-0.153	0.153	0.627	1.000	-0.503	0.561	0.151	-0.124
Class concentration	0.213	0.106	-0.106	-0.200	-0.503	1.000	-0.249	-0.110	0.130
Log assets	-0.097	-0.224	0.224	0.414	0.561	-0.249	1.000	0.140	-0.133
Leverage	-0.245	0.125	-0.125	0.037	0.151	-0.110	0.140	1.000	-0.364
Cash ratio	0.357	-0.298	0.298	0.044	-0.124	0.130	-0.133	-0.364	1.000
R&D intensity	0.340	-0.396	0.396	0.053	-0.051	0.112	-0.205	-0.315	0.513

Correlation matrix for the main numeric variables.

H Appendix H: Yearly Audit

Period	V1	V2	V3	V4
1991.0000	233.0000	0.6696	1.7221	320.5064
1992.0000	237.0000	0.6653	1.7417	325.8270
1993.0000	238.0000	0.6760	1.7913	337.9874
1994.0000	239.0000	0.6778	1.7348	349.7071
1995.0000	236.0000	0.6830	2.0166	366.8051
1996.0000	236.0000	0.6594	2.1379	380.8602
1997.0000	233.0000	0.6489	2.4754	404.3605
1998.0000	221.0000	0.6501	2.6527	440.2896
1999.0000	211.0000	0.6509	2.9393	488.1280
2000.0000	221.0000	0.6796	2.9787	519.8326
2001.0000	238.0000	0.6826	2.4734	543.5042
2002.0000	237.0000	0.6895	1.9995	526.8692
2005.0000	227.0000	0.6956	2.2091	268.2907
2006.0000	220.0000	0.6876	2.2944	280.4364
2007.0000	210.0000	0.6890	2.3005	374.0714
2008.0000	214.0000	0.6862	1.7092	516.5421
2009.0000	225.0000	0.6863	1.8849	640.5956
2010.0000	227.0000	0.6709	2.0194	736.5771
2011.0000	225.0000	0.6578	1.9323	811.2578
2012.0000	220.0000	0.6524	1.9836	890.6409
2013.0000	223.0000	0.6351	2.2694	974.1121
2014.0000	217.0000	0.6389	2.5169	1106.1613
2015.0000	204.0000	0.6297	2.4654	1249.4461

Year-by-year sample audit.

I Appendix I: Grouped Outcome Patterns

Quantile	N	MeanRegressor	MeanOutcome
5.0000	1039.0000	0.8832	1.9569
4.0000	1038.0000	0.7780	1.9779
3.0000	1038.0000	0.6925	1.9966
2.0000	1038.0000	0.5863	2.0965
1.0000	1039.0000	0.4011	2.8572

Quantile summary for the main regressor.

J Appendix J: Full Coefficient Table

term	estimate	std_error	statistic	p_value	model
tech_diff	-0.1935	0.6974	-0.2774	0.7841	arts_fe
log_assets	-0.5161	0.1442	-3.5785	0.0017	arts_fe
leverage	-0.6522	0.2840	-2.2967	0.0315	arts_fe
cash_ratio	1.5652	0.6384	2.4517	0.0226	arts_fe
rd_intensity	4.2903	1.5054	2.8500	0.0093	arts_fe
log1p(portfolio_patents)	-0.2660	0.0764	-3.4800	0.0021	arts_fe
class_count	0.0043	0.0024	1.8033	0.0851	arts_fe
class_hhi	0.4918	0.2947	1.6685	0.1094	arts_fe

Full coefficient dump exported from the local estimation script.

K Appendix K: Missingness Audit

Variable	Missing	MissingShare
Winsorized Tobin's Q	0.0000	0.0000
Technology differentiation	0.0000	0.0000
Neighbor similarity	0.0000	0.0000
Portfolio patents	0.0000	0.0000
Class count	0.0000	0.0000
Class concentration	0.0000	0.0000
Log assets	0.0000	0.0000
Leverage	0.0000	0.0000
Cash ratio	0.0000	0.0000
R&D intensity	507.0000	0.0977

Missingness audit for the variables summarized in the manuscript.

L Appendix L: Coverage of the Most Observed Entities

Entity	Observations	FirstPeriod	LastPeriod
10016.0000	23.0000	1991.0000	2015.0000
10405.0000	23.0000	1991.0000	2015.0000
10499.0000	23.0000	1991.0000	2015.0000
10519.0000	23.0000	1991.0000	2015.0000
1078.0000	23.0000	1991.0000	2015.0000
10983.0000	23.0000	1991.0000	2015.0000
10984.0000	23.0000	1991.0000	2015.0000
11456.0000	23.0000	1991.0000	2015.0000
11465.0000	23.0000	1991.0000	2015.0000
1161.0000	23.0000	1991.0000	2015.0000
11636.0000	23.0000	1991.0000	2015.0000
1209.0000	23.0000	1991.0000	2015.0000
1300.0000	23.0000	1991.0000	2015.0000
1356.0000	23.0000	1991.0000	2015.0000
1690.0000	23.0000	1991.0000	2015.0000
1722.0000	23.0000	1991.0000	2015.0000
1913.0000	23.0000	1991.0000	2015.0000
1988.0000	23.0000	1991.0000	2015.0000
2044.0000	23.0000	1991.0000	2015.0000
2086.0000	23.0000	1991.0000	2015.0000

Top entities by local observation count.

M Appendix M: Claim-Evidence Audit

Claim	EvidenceType	EvidenceFile	AuditNote
The source paper studies: The source paper asks whether technologically differentiated firms perform better than their peers.	Direct source fact	patent literature/arts-et-al-2023-position-and-differentiation-of-firms-in-technology-space.txt	Recovered from the source article before local writing begins.
The local executable panel contains 5,192 observations across 443 entities.	Local calculated fact	patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_analysis_dataset.csv — patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_sample_summary.csv	Backed by the verified analysis panel and sample audit.
The local headline estimate is -0.1935 for tech_diff with p = 0.7841.	Local calculated fact	patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_model_coefficients.csv	Backed by the exported coefficient file used in the paper.

Claim	EvidenceType	EvidenceFile	AuditNote
The manuscript uses local substitutions or analogues wherever the original data architecture is not fully recoverable.	Replication decision	Source-vs-decision table and limitations section	Used only when the original pipeline cannot be matched byte-for-byte from current local files.

Expanded claim-evidence audit.

N Appendix N: Source Inventory Note

Source Inventory - paper: Arts, Cassiman, and Hou (2023) - source title: Position and Differentiation of Firms in Technology Space - source file: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/arts-et-al-2023-position-and-differentiation-of-firms-in-technology-space.txt‘ ## Paper-First Reading Protocol - Step 1: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/arts-et-al-2023-position-and-differentiation-of-firms-in-technology-space.txt‘ — needed for: Read the original paper before touching the local estimates. — exists: n/a — status: workflow step - Step 2: ‘Position and Differentiation of Firms in Technology Space‘ — needed for: Extract the source paper’s theoretical logic, sample definition, variable definitions, and estimand. — exists: n/a — status: workflow step - Step 3: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_analysis_dataset.cs — C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/20 04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_model_coefficients.csv — C:/Users/bai/OneDriv - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_sample_summary.csv‘ — needed for: Match each required empirical object to a local executable file and verify that the file exists. — exists: n/a — status: workflow step - Step 4: ‘The manuscript, markdown notes, and PDF are generated from the verified file set.‘ — needed for: Write the replication paper only from the checked local files and calculated outputs. — exists: n/a — status: workflow step ## Local files - Source article: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/arts-et-al-2023-position-and-differentiation-of-firms-in-technology-space.txt‘ — needed for: Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices — exists: Yes — status: Direct source - Analysis panel: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_ — needed for: Estimate the local model and regenerate descriptive statistics — exists: Yes — status: Executable local dataset - Coefficient export: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_model_coefficients.csv‘ — needed for: Recover the re-

ported local coefficients and p-values — exists: Yes — status: Executable local result file - Sample summary: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_sample_summary.c — needed for: Audit sample construction and raw grouped moments — exists: Yes — status: Executable local result file - Fit summary: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_20 — needed for: Check model diagnostics when a fit file is available — exists: Yes — status: Executable local result file

O Appendix O: Study Map Note

Study Map - research question: The source paper asks whether technologically differentiated firms perform better than their peers. - positioning: The replication matters because it tests whether a central strategy result about technological differentiation survives when the semantic layer is stripped back to a fully executable local panel. - contribution: The paper's contribution is a transparent calculated analogue: it preserves the peer-comparison logic and performance outcome while making the measurement substitution explicit. - local design: The local rerun replaces the paper's text-semantic portfolio construction with a class-based five-year rolling USPC portfolio built from processed patent-firm assignments. - estimand: A firm and year fixed-effects regression of winsorized Tobin's Q on local technology differentiation and financial controls. - headline estimate: '-0.1935' with 'p = 0.7841'. ## Decision Audit - Research question: The source paper asks whether technologically differentiated firms perform better than their peers. - Unit of analysis: Firm-year - Sample frame: Processed public-firm GVKEY-year panel with patent portfolios and financial controls. - Time window: 1991 to 2015 in the local executable panel. - Outcome definition: Winsorized Tobin's Q at the firm-year level. - Main explanatory design: Firm and year fixed-effects regression of Tobin's Q on local technology differentiation and controls. - Controls and fixed effects: Financial controls plus firm and year fixed effects. - Exclusion or substitution logic: Original semantic embedding inputs are unavailable locally, so the executable rerun substitutes class-based portfolio similarity and keeps only observations supported by the processed panel. ## Source Versus Decision - direct source fact: The source paper uses semantic patent content. - direct source fact: Tobin's Q is a central outcome. - direct source fact: Differentiation is relative to peers. local replication decision: The local rerun uses processed primary-USPC class portfolios. local replication decision: The local panel keeps Tobin's Q because it is fully reconstructible. local replication decision: Peer comparison is preserved with a class-based similarity measure.

P Appendix P: Data Design Note

Data Design - DISCERN-linked firm-patent assignments with unambiguous GVKEYs. - PatentView-derived utility-patent and primary-USPC class information. - Compustat annual fundamentals used to reconstruct Tobin's Q and controls. - panel file used: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_analysis_dataset.csv‘ - coefficient file used: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_model_coefficients.csv‘ - sample summary used: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_sample_summary.csv‘ - observations: ‘5,192‘ - entities: ‘443‘ ## Required Data And Local File Mapping - Source article: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/arts-et-al-2023-position-and-differentiation-of-firms-in-technology-space.txt‘ — needed for: Recover theory, hypotheses, sample rules, definitions, tables, notes, and appendices — exists: Yes — status: Direct source - Analysis panel: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_ — needed for: Estimate the local model and regenerate descriptive statistics — exists: Yes — status: Executable local dataset - Coefficient export: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_model_coefficients.csv‘ — needed for: Recover the reported local coefficients and p-values — exists: Yes — status: Executable local result file - Sample summary: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_sample_summary.c — needed for: Audit sample construction and raw grouped moments — exists: Yes — status: Executable local result file - Fit summary: ‘C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_20 — needed for: Check model diagnostics when a fit file is available — exists: Yes — status: Executable local result file

Q Appendix Q: Derivations Note

Derivations And Estimands - The local headline coefficient on technology differentiation is negative and statistically weak after firm and year effects are imposed. - A firm and year fixed-effects regression of winsorized Tobin's Q on local technology differentiation and financial controls. - model equations are reproduced in ‘paper/main.tex‘. ## Variable Map - Technology portfolio -¿ ‘portfolio_patents; class_count; class_hhi‘ : Aggregate utility patents by GVKEY and primary USPC class over t-5 to t-1. - Technology differentiation -¿ ‘tech_diff‘ : Compute one minus average similarity to close peers. - Peer benchmark -¿ ‘neighbor_similarity‘ : Use local class-based portfolios to define the similarity set. - Firm performance -¿ ‘tobin_q_w‘ : Reconstruct Tobin's Q from processed Compustat variables and winsorize. - Financial controls -¿ ‘log_assets; leverage; cash_ratio; rd_intensity; log1p(portfolio_patents)‘ : Carry forward the financial controls already built in the local script. ## Claim Evidence Audit - Direct source fact: The source paper studies: The source paper asks whether technologically differentiated firms perform better than their peers. — evidence: patent literature/arts-et-al-2023-position-and-differentiation-of-firms-in-technology-space.txt - Local calculated fact: The local executable panel contains 5,192 observations across 443 entities. — evidence: patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_analysis_dataset.csv — patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_sample_summary.csv - Local calculated fact: The local headline estimate is -0.1935 for tech_diff with p = 0.7841. — evidence: patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_model_coefficients. - Replication decision: The manuscript uses local substitutions or analogues wherever the original data architecture is not fully recoverable. — evidence: Source-vs-decision table and limitations section

R Appendix R: Replication Decisions Note

Replication Decisions - The exact text-embedding pipeline from the source paper is not stored locally. - The local differentiation measure is therefore class-based rather than text-semantic. - Comparisons to the published estimates should be read as inferential rather than byte-identical. - direct source facts are separated from local replication decisions in the manuscript. - no coefficient or table is fabricated. ## Source Facts Versus Local Decisions - source: The source paper uses semantic patent content. - source: Tobin's Q is a central outcome. - source: Differentiation is relative to peers. decision: The local rerun uses processed primary-USPC class portfolios. decision: The local panel keeps Tobin's Q because it is fully reconstructible. decision: Peer comparison is preserved with a class-based similarity measure.

S Appendix S: Local Execution Status Note

Local Execution Status - batch date: ‘2026-04-04‘ - status: long-form journal-style manuscript generated from local result files - current paper: ‘arts-cassiman-hou-2023-technology-space‘

T Appendix T: Cleaned Output Tables Note

Cleaned Output Tables - paper-first reading protocol - required-data and file-manifest table - decision register - sample reconstruction table - local source inventory table - source facts versus replication decisions table - claim-evidence audit table - variable mapping table - key coefficient table - full coefficient table - descriptive statistics - correlation matrix - yearly audit - missingness audit - top-entity coverage table - quantile summary

U Appendix U: Build Notes

Build Notes

- batch date: ‘2026-04-04‘ - manuscript style: long-form journal-style replication draft - evidence rule: only processed local datasets and actual local estimates are used - workflow rule: read the source paper first, then map required datasets to verified local files, then write the paper from those files - skill path guiding the structure: ‘C:/Users/bai/.codex/skills/write-replication-journal-github/SKILL.md‘

V Appendix V: Manuscript Metadata

Field	Value
Paper	Arts, Cassiman, and Hou (2023)
Source title	Position and Differentiation of Firms in Technology Space
Authors	Sam Arts, Bruno Cassiman, and Jianan Hou
Outlet	Management Science
Batch date	2026-04-04
Rows	5,192
Entities	443
Headline term	tech_diff
Headline estimate	-0.1935
Headline p-value	0.7841
Panel path	C:/Users/bai/OneDrive - Tilburg University/Desktop/replicate Ambions/patent literature/daily_replications/2026-04-04/arts-cassiman-hou-2023-technology-space/results/arts_2023_analysis_dataset.csv

Metadata block for GitHub publishing and future reruns.

W Appendix W: Replication Summary Note

Research question: The source paper asks whether technologically differentiated firms perform better than their peers. Local design: The local rerun replaces the paper's text-semantic portfolio construction with a class-based five-year rolling USPC portfolio built from processed patent-firm assignments. Estimand: A firm and year fixed-effects regression of winsorized Tobin's Q on local technology differentiation and financial controls. Local result focus: The local headline coefficient on technology differentiation is negative and statistically weak after firm and year effects are imposed. Comparison note: The source paper reports a positive performance relationship, whereas the local executable analogue produces a negative and imprecise coefficient. That gap is useful evidence about how much the published result may depend on the original semantic measurement layer and sample construction.

References

[1] Sam Arts, Bruno Cassiman, and Jianan Hou. Position and differentiation of firms in technology space. *Management Science*, 2023.